

Recommendation System of News Based on Impression Logs

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1 Abstract

This paper aims to enhance new recommendation systems by leveraging the MIND dataset to capture and analyze user-article interactions. The MIND dataset is a large-scale dataset of news articles and user behavior logs that enable the study of user preferences and news consumption patterns. The proposed approach integrates a GRU module into the DKN framework to capture the temporal dynamics of user news browsing history and improve personalized news recommendations [1][2]. The analysis compares the distribution of news types in user engagement versus news content exposure. It uses the Jensen-Shannon divergence to quantify the alignment between user preferences and the recommender’s inferences. The key findings identify areas where the recommendation algorithm may be over or underexposing certain news types to users and highlight opportunities to improve the alignment between user preferences and recommended content. The insights can inform strategies to enhance news recommendation systems and serve user interests better.

Contributors: Jure

2 Introduction

The MIND: Microsoft News Dataset, a comprehensive repository comprising news articles and user behavior logs, presents a valuable resource for studying user-article interactions and refining news recommendation systems. This paper seeks to harness the rich information within the MIND dataset to gain insights into user preferences and news consumption patterns, thereby informing the development of strategies to enhance news recommendations.

The DKN (Deep Knowledge-Aware Network) [2] represents a sophisticated framework for news recommendation, integrating knowledge graph information to enrich the representation of news articles. However, while DKN effectively leverages semantic and entity-level knowledge, it does not explicitly model the temporal dynamics of user news browsing history.

GRUs (Gated Recurrent Units) are a type of recurrent neural network particularly adept at capturing long-term dependencies in sequential data, such as users’ news consumption patterns over time. Integrating a GRU module into the DKN framework holds the potential to enhance the model’s understanding of the temporal aspects of user preferences, thereby facilitating improved personalized news recommendations [1].

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2.1 Motivation

This paper endeavors to enrich news recommendation systems by leveraging the MIND dataset to capture and analyze authentic user-article interactions. News recommendation is pivotal in facilitating user exploration of relevant content and enabling news providers to engage their audience effectively. However, prevailing recommendation approaches often struggle to capture users’

nuanced preferences and evolving interests comprehensively. Our motivation stems from recognizing potential biases within news recommendation algorithms, prompting us to explore this critical area.

Specifically, our paper addresses the following key aspects: diversity, fairness, and impact on user behavior. For diversity, we assess whether news recommendation algorithms effectively provide diverse content to expose users to varied perspectives. For fairness, we examine the presence of diversity in news consumption to mitigate the formation of filter bubbles. To assess the impact on user behavior, we investigate how news recommendations influence user news consumption patterns and opinions.

Our motivation is to identify biases within news recommendation algorithms to comprehend the exposure to specific topics. Subsequently, we aim to explore methods for mitigating these biases within the algorithms. Investigating biases in news recommendation algorithms is essential to ensure the delivery of high-quality and diverse news experiences to users.

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2.2 Related Work

The literature concerning bias in recommendation systems, particularly within news recommendation algorithms, underscores the significance of addressing biases in news content representations. Prior research has identified various types of biases in recommendation algorithms, including algorithmic bias, personalization bias, and content bias. These biases can contribute to the formation of filter bubbles, echo chambers, and the amplification of misinformation, thereby significantly influencing user experiences. Diversity analysis has emerged as a prevalent method for identifying biases within recommendation systems. Fairness-aware recommendation frameworks have also been proposed to mitigate biases and ensure equitable content distribution across user demographics. Our research focuses on identifying bias within news recommendation algorithms utilizing the MIND database, renowned for its extensive collection of behavior logs from the Microsoft News website. Leveraging the DKN and GRU approaches, we employ content filtering to enhance the diversity and fairness of recommendations.

The DKN represents a sophisticated news recommendation framework integrating a knowledge graph into its architecture [2]. The process begins by ingesting a candidate news article alongside news articles previously accessed by the user. Subsequently, the news headlines transform a KCNN (Knowledge-aware Convolutional Neural Network), adept at assimilating symbolic knowledge from the knowledge graph to enhance the learning of sentence representations. While DKN effectively utilizes semantic and entity-level knowledge, it does not explicitly model the temporal dynamics of user news browsing history. GRUs, a recurrent neural network well-suited for capturing long-term dependencies in sequential data, offer a solution for modeling a user’s news consumption patterns over time. Integrating a GRU module into the DKN framework holds the potential to enhance the model’s

understanding of the temporal aspects of user preferences and, consequently, improve personalized news recommendations [1].

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3 Formal Problem Statement

The primary objective of this paper is to investigate biases present within news recommendation algorithms. Specifically, we aimed to identify instances of overexposure and underexposure to certain topics and develop effective methods for mitigating these biases.

To achieve this objective, we utilized the MIND: Microsoft News Dataset, a widely employed database that contains behavior logs from the Microsoft News website. This dataset, known for its comprehensive and diverse user behavior data, is instrumental in our research as it allows us to analyze human behavioral patterns and the exposure of various topics, thereby enhancing our understanding of the underlying dynamics within news recommendation algorithms.

The paper aims to enhance news recommendation systems by leveraging the MIND dataset to capture and analyze genuine user-article interactions. We propose integrating a GRU module into the DKN framework better to capture the temporal dynamics of user news browsing history and improve personalized news recommendations. The key aspects of the problem statement include: Utilizing the MIND dataset, a valuable resource for studying user-article interactions and developing enhanced news recommendation systems. Enhancing the DKN news recommendation framework by incorporating a GRU module to model the sequential patterns and dependencies in a user’s news consumption over time [1][2]. Improving the quality of personalized news recommendations by leveraging the temporal information captured by the GRU to provide a richer and more nuanced representation of user interests [1] and analyzing the alignment between user preferences and the recommender’s inferences by comparing the distributions of news types in user engagement versus news content exposure, using metrics like Jensen-Shannon divergence. Identify areas where the recommendation algorithm may be over- or under-exposing certain news types to users and explore opportunities to better align recommended content with user interests.

By addressing these key aspects, the paper aims to enhance news recommendation systems and better serve user needs. The insights attained from our research can potentially lead to more accurate and personalized news recommendations, thereby improving user satisfaction and engagement.

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4 Dataset

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4.1 Microsoft News Dataset

The Microsoft News Dataset (MIND) [3] represents an extensive compilation of English-language news assembled from anonymized user behavior logs collected via the Microsoft News website. Its creation was designed to facilitate the enhancement of news recommendation systems by enabling the capture, analysis, and utilization of genuine user-article interactions. The MIND dataset comprises 1,000,000 user behavior logs, 161,013 news articles, and 15,777,377 impression records.

The dataset consists of four primary files: **behaviors**, **news**, **entity_embedding**, and **relation_embedding**. The **behaviors**

file logs user interactions, including Impression ID, User ID, Time, History, and Impressions. This file tracks user clicks on news articles and provides both historical and current impressions with timestamps. The **news** file contains comprehensive metadata for news articles, such as News ID, Category, SubCategory, Title, Abstract, and URL. Additionally, it features entities from titles and abstracts paired with their respective Wikidata IDs. Tables 1 and 2 demonstrate examples of news records and user logs in MIND, respectively. Both **entity_embedding** and **relation_embedding** files house 100-dimensional embeddings of news entities and relations. These embeddings are sourced from the WikiData knowledge graph.

Column	Content
News ID	N37378
Category	sports
SubCategory	golf
Title	PGA Tour winners
Abstract	A gallery of recent winners on the PGA Tour.
URL	https://www.msn.com/en-us/sports/golf/pga-tour-winners/ss-AAjnQjj?ocid=chopendata
Title Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [0], "SurfaceForms": ["PGA Tour"]}]
Abstract Entities	[{"Label": "PGA Tour", "Type": "O", "WikidataId": "Q910409", "Confidence": 1.0, "OccurrenceOffsets": [35], "SurfaceForms": ["PGA Tour"]}]

Table 1: Example of a news record in MIND

Column	Content
Impression ID	91
User ID	U397059
Time	11/15/2019 10:22:32 AM
History	N106403 N71977 N97080 N102132 N97212 N121652
Impressions	N129416-0 N26703-1 N120089-1 N53018-0 N89764-0 N91737-0 N29160-0

Table 2: Example of a user log in MIND

The MIND dataset offers insights into user-article interactions, detailed article metadata, and embeddings suitable for knowledge-driven recommendations.

4.2 MIND-demo

Due to computational limitations in our experiments, we have opted to use the MIND-demo dataset, a subset of the larger MIND dataset. Despite its reduced size, the MIND demo maintains the same structure as the full MIND dataset, offering a scaled-down version suitable for preliminary research and development. The MIND-demo dataset includes behavior logs from 500 users, 9,432 news articles, and 6,134 impression logs, providing a smaller yet representative sample of the data available in the full MIND dataset.

By leveraging the MIND-demo dataset in our experiments, we aim to simulate realistic news recommendation scenarios in the real world using limited computational resources.

5 Proposed Approach

5.1 Deep Knowledge-Aware Network

The Deep Knowledge-Aware Network (DKN) [2] is a sophisticated news recommendation framework that effectively integrates the concept of a knowledge graph into its underlying architecture. It commences by ingesting a candidate news article along with news articles previously accessed by the user. This process transforms the news headlines into a Knowledge-aware Convolutional Neural Network (KCNN). This specialized KCNN is adept at assimilating symbolic knowledge from the knowledge graph to enhance the learning of sentence representations.

Knowledge distillation techniques are initially applied to better understand and distinguish entities within the news context. This allows for identifying pivotal entities, which are subsequently employed to formulate a sub-graph. The insights derived from this distilled knowledge are then leveraged to augment the representation of the news content. This enriched representation is combined with the original embeddings and supplied as input to the KCNN. As a result, the KCNN produces an integrated embedding vector that captures both the semantic essence and the extracted knowledge from the knowledge graph.

An attention mechanism is deployed to discern the varying influences of a user’s historical preferences on the candidate news. This attention network dynamically computes the user’s aggregated historical profile, thereby refining the discernment of user preferences. Lastly, a Multi-Layer Perceptron (MLP) is employed to ascertain the probability of a user clicking on the candidate news article. This prediction is based on their historical interactions and the enriched embedding generated by the KCNN.

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5.2 Gated Recurrent Unit

Gated Recurrent Units (GRUs) [1] stand out as recurrent neural networks tailored to manage long-term dependencies in sequential data, such as a user’s news browsing history.

GRUs integrate three pivotal mechanisms at their core: the update gate, reset gate, and candidate activation function. The update gate governs the retention of the previous state, deciding which information is relevant to carry forward. In contrast, the reset gate allows for the selective refreshing of outdated information, ensuring the model focuses on the most pertinent inputs. The candidate activation function synthesizes the current input with the retained or updated previous state to compute a new state.

With these interconnected features, GRUs maintain a contextual memory across sequences, enabling them to capture intricate sequential patterns and dependencies. This capability equips them to discern complex behaviors and preferences in a user’s news consumption, providing a nuanced understanding of evolving user interactions over time.

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5.3 Proposed Modification

Our proposed modification to the DKN model involves integrating a GRU module to capture the temporal dynamics inherent in a user’s news browsing history. After obtaining the presentation of a user’s news browsing history from the DKN model, this time-series data is sequentially processed through the GRU module.

The GRU captures the sequential patterns and dependencies in the user’s interaction with news articles over time. To syn-

ergize this temporal information with the enriched embedding generated by the KCNN in the DKN model, the final output of the GRU is concatenated with the DKN’s enriched embedding vector. This concatenated representation is then fed into the MLP for click-through rate prediction.

By integrating the GRU module into the DKN framework after the knowledge distillation and KCNN processes, our objective is to enhance the model’s ability to capture and leverage the temporal dynamics present in a user’s news browsing history. This enhancement aims to provide a richer and more nuanced representation of the user’s interests, thereby improving the quality of personalized news recommendations. Figure 1 illustrates the overall structure of our proposed approach.

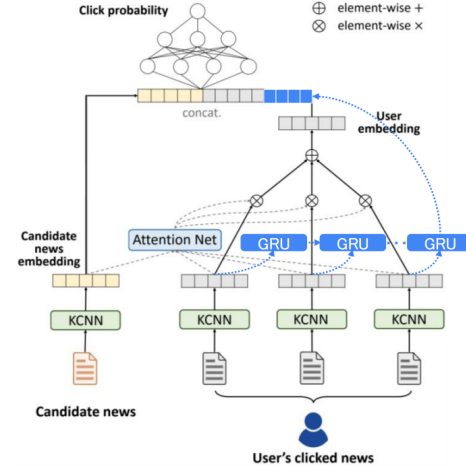


Figure 1: The network structure of our proposed approach

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6 Results and Observations

6.1 Comparing News Type Distributions

The MIND dataset provides a **Type** label for each news article, which can be leveraged to categorize news articles into different groups. The discrepancy in the distribution of news types in the user engagement and the recommender-ranked news content exposure serves as a crucial indicator for identifying if there is significant overexposure or underexposure of certain types of news to each user.

Building a distribution of news types in users’ actual engagement is straightforward. For each user, we first identify the type of each news article. As mentioned before, this information is readily accessible in the impression logs. Then, we count the occurrence of different news types of the user’s clicked-on news articles across the impression log.

Extracting a news type distribution from the exposure is more tricky. Simply counting the occurrence of each news type over the impression log does not make general sense. This is because the recommendation algorithm is not responsible for generating the exact content to be offered to the users. Instead, it produces a ranking of all the news articles presented in the impression log of each user, with a lowest rank of 1. News articles with higher rankings are more likely to be clicked by the user than those with lower rankings.

To effectively incorporate the importance placed on each news type by the recommendation algorithm, We propose an inverse-rank weighting strategy for counting the occurrence of each news

in the impression. Specifically, within the impression log of a specific user, we assign an inverse-rank weighted count for each news article, which is formulated as the total number of news articles in the impression log divided by the rank of this news article. In this way, the types of news articles that are deemed to highly align with the user’s preference by the recommendation algorithm will be assigned more counts. With this, we can build a distribution of news types of impressions with aggregated information from the recommendation algorithm.

In the following sections, we first analyze the relationship between the types of news items users are exposed to versus the types they click on. We present an example of comparing news type distributions of user engagement to news content exposures for individual users. We will then use a statistical metric to evaluate discrepancies in the expected versus actual user interactions with news content.

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6.1.1 Individual Exposure versus Clicks

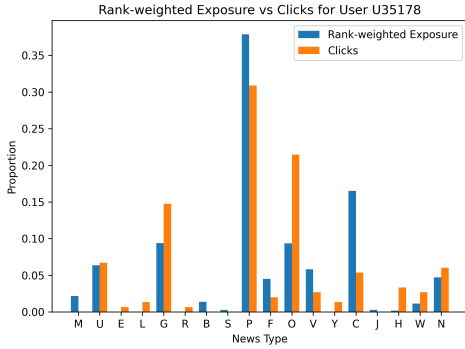


Figure 2. Exposure vs clicks for user35178.

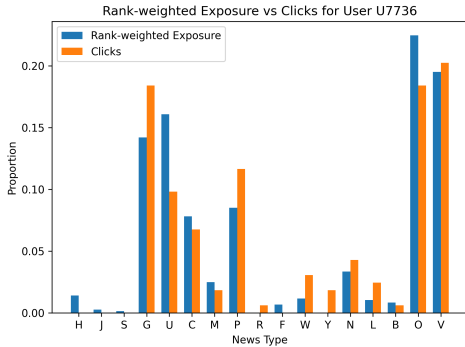


Figure 3. Exposure vs clicks for user7736.

Figure 2 and Figure 3 each present a comparison of the portion of news types exposures to clicks for a particular user. As shown in Figure 2, based on the inference from the click behavior of user35178, the top 4 news types of this user’s actual preference are P, O, G, C, respectively. Meanwhile, according to the ranking results by the recommendation algorithm for this user, the top 4 news types align with the user preference. However, each has some extent of underexposure or overexposure. For example, the proportion of type O in exposure is less than half of that in real user engagement, forming a situation of underexposure of type O, and the proportion of type C in the exposure is over twice of that in user engagement, which leads to overexposure of type C for user35178. Figure 3 illustrates a case where news type distributions from news content exposure and actual user engagement highly align with each other. However, there is a misalignment in certain types; for instance, in type G with a deviation of 16%

and type U with 50%, the recommender maintains a high degree of alignment for user7736.

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6.1.2 Measure of similarity: Jensen-Shannon Divergence

In addition to the straightforward demonstration, we perform a quantitative analysis to measure the divergence between two news type distributions: one representing the weighted exposure of news types a user has seen and another representing the actual news types on which the user has clicked. Specifically, we use Jensen-Shannon divergence (JSD) as the divergence metric. JSD values range from 0 to 1, where 0 indicates perfect alignment, and 1 indicates maximum possible divergence. Kullback-Leibler divergence can be infinite if two distributions are not on the same support, while JSD is beneficial for measuring the similarity between two distributions as it is always finite and symmetrical. By analyzing the density of JSD values, we can identify whether the recommendation algorithm correctly predicts users’ preferences or if there are discrepancies (higher JSD values). A peak near lower JSD values in the plot would indicate effective alignment between user preferences and the inference made by the recommendation algorithm. Conversely, a spread or shift towards higher values might suggest areas for improvement in the recommendation algorithms or content delivery strategies.

We calculate the JSD for each user based on their news impressions and clicks and visualize this result across all users in the form of a kernel density estimate plot. This showcases the variability and density of JSD values among all users, indicating how their news viewing habits differ from the recommender-ranked exposure of news content.

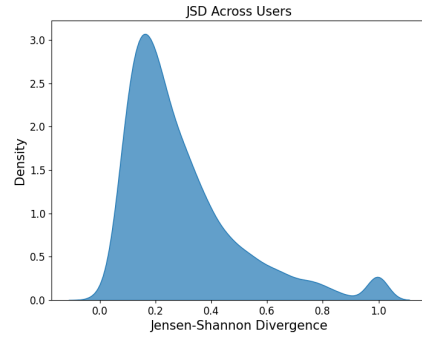


Figure 4. JS Divergence.

As shown in Figure 4, the JS divergence values across all users predominantly cluster around 0.2. This indicates a moderate divergence between the two probability distributions being compared, suggesting that while there is some alignment between what users interact with and the intended content exposure by the recommendation algorithm, notable discrepancies still exist.

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6.2 Diversity Analysis

Measuring the diversity of the ranked impressions suggested by the recommendation algorithm helps ensure that the recommender neither narrows the users’ content exposure too much, which might limit their engagement nor overwhelms them with too much breadth, which might dilute personal relevance. We use two statistical tools, the Gini coefficient and entropy, to assess the diversity of recommendations for each user, providing insights

into how well a recommendation system can provide users with a wide range of content.

6.2.1 Gini Coefficient

The Gini Coefficient is a measure traditionally used in economics to gauge income inequality but can be effectively adapted to measure the inequality in exposure of different news types in recommendation systems. In this context, it helps assess whether a system disproportionately favors certain types of news over others.

A lower Gini Coefficient might be desirable to ensure that users are exposed to a broad array of topics, preventing the filter bubble effect, in which a user sees a very narrow range of content.



Figure 5. Distribution of Gini coefficients across users.

As shown in Figure 5, the Gini coefficients across all users cluster at a value of 0.5, suggesting a moderate diversity level. This indicates some inequality in how news types are distributed among users’ recommendations. The value of 0.5 is at the midpoint between perfect equality and maximal inequality; it suggests that while some news types may be recommended more frequently than others (as intended by the recommendation algorithm), the disparity is not extremely pronounced.

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6.2.2 Entropy

Entropy measures the uncertainty or randomness in a system. Applied to news recommendations, it quantifies how unpredictable the distribution of news types is across the recommendations made to users. Specifically, we calculate each user’s entropy of ranked exposure using the probability distribution of news types across ranked impressions. The entropy is maximized if all news types are equally likely, indicating maximum diversity. Lower entropy indicates that some news types are more prevalent in the recommendations than others, suggesting predictability or bias in content delivery.

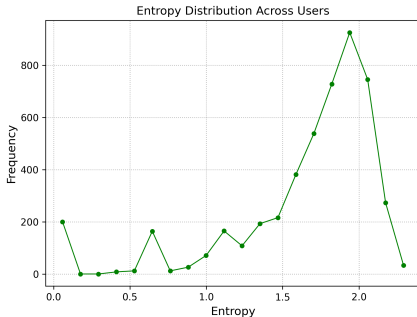


Figure 6. Distribution of entropy across users.

Figure 6 illustrates the distribution of the entropy of news types in ranked exposure out of user impressions across all users.

There are 22 news types available. An entropy of 2.0, while not the maximum for 22 categories (which would be about 4.46), still indicates a moderate level of diversity in news types in the ranked exposure for users.

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7 Conclusion

We employ a variety of statistical tools to assess how different news types are represented as intended by the recommendation algorithm and interacted with by users. Our methods included Gini coefficients, entropy measurements, and Jensen-Shannon divergence, each providing unique insights into the diversity of content exposure and user engagement. While the algorithm performs adequately in presenting a variety of news types to users, there is still evident potential for fine-tuning the accuracy of content exposure to ensure close alignment with user preferences.

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7.1 Future Work

Future work could explore additional techniques to enhance the news recommendation model, such as incorporating more advanced temporal modeling methods or leveraging external knowledge sources to enrich the representation of news articles. Additionally, conducting user studies could provide valuable insights into how users perceive and interact with the improved news recommendations. The goal of continuing to refine the recommendation approach and acquire a deeper understanding of user preferences is to develop news recommendation systems that better serve users’ evolving information needs.

Several avenues for future research are achieving algorithmic fairness and diversity in news recommendation systems, while these aspects are critical for enhancing the inclusivity and quality of news recommendations. Another is exploring the influence of human actions on the spread of news. However, investigating this aspect was within the scope of our research interests, and delving deeper into this area required suitable algorithms, so we could not pursue this aspect of the research as extensively. Also, when conducting experiments on a larger dataset and comparing the results, testing our model on a larger dataset and comparing the results with those obtained from the smaller dataset would be beneficial. This comparison would help determine the generalizability and robustness of our approach.

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